DIPLOMA PROGRAMME PRACTICE · SL · SET B

Chemistry / Paper 1A + 1B

Worked mark scheme · 55 marks · 90 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer C Gas-to-solid deposition of I_2 preserves its molecular identity. NH_3 and HCl react chemically to produce NH_4Cl.

PAPER 1A · QUESTION 2 [1]

Answer D There are 24 protons, so this is a 3+ ion. Adding three electrons neutralizes it without changing the nucleus.

PAPER 1A · QUESTION 3 [1]

Answer C Equal atom counts give a combined mean of 20.6. The heavy fraction is (20.6−20)/2=0.30.

PAPER 1A · QUESTION 4 [1]

Answer D The 2p configurations are $2p^3$, $2p^4$, $2p^5$ and $2p^6$, giving 3, 2, 1 and 0 unpaired electrons by Hund’s rule.

PAPER 1A · QUESTION 5 [1]

Answer A Each mole of $CaCl_2$ provides two moles of Cl^-. Required stock volume is 0.00600/0.200=0.0300 dm^3.

PAPER 1A · QUESTION 6 [1]

Answer A At fixed T and p, density is proportional to molar mass. The ratio is (0.400/100)/(0.300/150)=2.00.

PAPER 1A · QUESTION 7 [1]

Answer B At fixed V, n is proportional to p/T. n_2=0.600×(81/84)×(280/360)=0.450 mol.

PAPER 1A · QUESTION 8 [1]

Answer C Three Mg^{2+} and two N^{3-} balance charges. Displacement can align like charges, whose repulsion splits the lattice.

PAPER 1A · QUESTION 9 [1]

Answer B An O lone pair can accept a hydrogen bond. A pure sample also needs donors such as O–H for conventional intermolecular hydrogen bonding.

PAPER 1A · QUESTION 10 [1]

Answer C BF_3 is trigonal planar with three equivalent B–F bonds. The vector sum vanishes, unlike the other molecules.

PAPER 1A · QUESTION 11 [1]

Answer B Mobile delocalized electrons carry current through the solid metal. Ionic conduction requires motion of charged ions.

PAPER 1A · QUESTION 12 [1]

Answer A Each repeat has mass 42, so there are 1000 repeats. Two of the three carbons per repeat lie in the backbone.

PAPER 1A · QUESTION 13 [1]

Answer B X is sodium. Compared with magnesium, it has one fewer valence electron and a larger radius due to lower effective nuclear attraction.

PAPER 1A · QUESTION 14 [1]

Answer C Br^- supplies electrons to Cl_2. Each bromide ion loses one electron as elemental bromine forms.

PAPER 1A · QUESTION 15 [1]

Answer D 2-methylpropanal has four carbons and an aldehyde group. Butan-2-one has the same molecular formula but a ketone group.

PAPER 1A · QUESTION 16 [1]

Answer D Oxidation of secondary butan-2-ol gives butan-2-one, C_4H_8O, by removal of two hydrogen atoms.

PAPER 1A · QUESTION 17 [1]

Answer C Water gains 150×4.18×8.0=5016 J. Divide 5.016 kJ by 0.400 g.

PAPER 1A · QUESTION 18 [1]

Answer B Bonds broken require 436+243=679; formation releases 2×431=862 kJ. ΔH=679−862.

PAPER 1A · QUESTION 19 [1]

Answer D P gives 2.5/50=0.050 and Q gives 1.8/30=0.060 kg CO_2 per MJ of chemical energy.

PAPER 1A · QUESTION 20 [1]

Answer A $2CO+O_2 \rightarrow 2CO_2$. The 60 cm³ CO uses 30 cm³ O_2 and yields 60 cm³ CO_2. Add 10 cm³ excess O_2.

PAPER 1A · QUESTION 21 [1]

Answer A The stoichiometric atom economy is 88/(60+46)×100=83.0%. Actual yield is a different measure.

PAPER 1A · QUESTION 22 [1]

Answer A $HCO_3^- + H^+ \rightarrow CO_2 + H_2O$ is 1:1. 0.0100 mol acid at 0.250 mol dm^{-3} occupies 0.0400 dm^3.

PAPER 1A · QUESTION 23 [1]

Answer A The initial tangent measures the rate at equal initial concentrations. Final volume is governed by limiting reactant.

PAPER 1A · QUESTION 24 [1]

Answer B Expansion favours two gas particles over one. Heating favours the endothermic reverse reaction.

PAPER 1A · QUESTION 25 [1]

Answer D Exchange of a label demonstrates continuing molecular conversion. Constant bulk concentrations arise because the two rates are equal.

PAPER 1A · QUESTION 26 [1]

Answer B Equal amounts of OH^- neutralize equal volumes, so analytical concentrations match. The higher pH corresponds to less hydrogen ion and less ionization.

PAPER 1A · QUESTION 27 [1]

Answer B Excess H^+=0.00100 mol in 0.0300 dm^3. pH=$-\log_{10}(0.0333)$=1.48.

PAPER 1A · QUESTION 28 [1]

Answer C $Zn \rightarrow Zn^{2+}+2e^-$ supplies enough electrons to reduce two Ag^+ per Zn atom.

PAPER 1A · QUESTION 29 [1]

Answer D Adding the two propagation equations cancels both radicals and regenerates Br• as a chain carrier.

PAPER 1A · QUESTION 30 [1]

Answer A Addition across a C=C bond consumes bromine without UV initiation. The other observations are also possible for saturated hydrocarbons.

PAPER 1B · QUESTION 1 [11]

Two solvents for a plant extract

(a) Calculate Rf values for V and Q in each run. [3]

1 In run I both V and Q give 5.6/8.0=0.70.

1 In run II V gives 3.0/6.0=0.50 and Q gives 4.2/6.0=0.70.

1 Rf is a distance ratio from the same baseline and has no units.

(b) Assess the identifications U=P and V=Q, using both runs. [3]

1 U matches P in both runs (0.25 and 0.30); this supports but does not prove identity.

1 V matches Q in run I but not in run II, evidence against V=Q.

1 Different compounds can co-migrate under one set of conditions; a co-run or independent technique gives stronger evidence.

(c) Within run I, compare how the phases affect P and Q. Explain why absolute migration distances alone cannot be compared between the two runs. [3]

1 P travels less relative to the front; it is more strongly retained by silica relative to its affinity for that mobile phase.

1 Q travels further and partitions relatively more into the mobile phase.

1 Front distances and solvent compositions differ between the runs; the migration balance and distance scale change.

(d) Predict the effect of submerging the starting spots. State one further practical precaution for obtaining reliable Rf values. [2]

1 Sample may dissolve into the solvent reservoir and lose its defined starting point.

1 Mark the front immediately on removal before evaporation; accept use of a pencil baseline, small spots or a level plate.

PAPER 1B · QUESTION 2 [14]

Standardizing a monoprotic acid

(a) Determine the three accurate titres and their mean. Explain the treatment of the rough trial. [3]

1 Accurate titres: 24.90, 24.95 and 24.90 cm^3.

1 Mean=24.9167 cm^3, reported as 24.92 cm^3.

1 The rough run locates the endpoint and is not included; the three accurate titres agree within 0.05 cm^3.

(b) Calculate the concentration of the sodium carbonate solution and the amount in each aliquot. [3]

1 n=1.325/106.0=0.01250 mol.

1 c=0.01250/0.2500=0.05000 mol dm^{-3}.

1 Aliquot contains 0.05000×0.02500=0.001250 mol.

(c) Write the ionic equation and determine the HCl concentration. [3]

1 $CO_3^{2-}+2H^+ \rightarrow CO_2+H_2O$.

1 The aliquot consumes 0.002500 mol HCl.

1 c=0.002500/0.0249167=0.1003 mol dm^{-3}.

(d) Estimate the maximum percentage uncertainty from the two burette readings for a single accurate titre. Explain why close titres do not rule out a calibration error. [3]

1 Add reading uncertainties to get ±0.10 cm^3 per titre.

1 0.10/24.92×100≈0.40%.

1 A consistent calibration offset in delivered volume can affect all runs similarly, so precision alone does not establish accuracy.

(e) Suppose the carbonate had absorbed water before weighing. Deduce the direction of error in the calculated acid concentration if the full mass is treated as anhydrous. [2]

1 The true moles of carbonate in the weighed sample would be lower.

1 The calculation assigns too much HCl to the measured titre, so the calculated HCl concentration is too high.